Capstone Project Proposal Report
(Individual Report)

Instructions:
This form is to be completed by each student doing Project registration to fulfill their senior design or capstone requirement. It must be completed and submitted to your Guide. Each student must complete this form individually.
This report is to be completed during the starting of the semester, while the project description report will be completed during end of the semester.

Guide Approval (initials/date):		

CAP4001– Capstone Project Proposal Report

Student Name	**Ameer Roshan Shaik**
Student Register Number	**23BCE7692**
Programme	**CSE AI/ML**
Semester/Year	**7th Semester**
Guide(s)	**Balusa Bhanu Chander**
Project Title	**An Explainable AI-Based Decision Support System for Dynamic Water Quality Index Prediction and Recommendation System**

Team Composition: Provide the information below for each member of the **project team**. Include **all** project team members, not just those in your discipline or those enrolled for Capstone project. Please also include yourself!

Reg. No	Name	Major	Specialization
23BCE9594	Chintakayala Karthikeya	CSE	AI/ML
23BCE7692	Ameer Roshan Shaik	CSE	AI/ML
23BCE9533	Shaik Abdul Samad	CSE	AI/ML
23BCE8722	Desam Pranay Kumar Reddy	CSE	AI/ML

Project and Task Description:

(a) Project Summary

Water quality assessment helps in providing the safe water, protecting the environment, and managing the water resources in an effective manner. Now the process of determining the Water Quality Index (WQI) is done manually by considering the number of water quality parameters. It is therefore not effective for large-scale implementations.

The current study aims to develop an Explainable Artificial Intelligence (XAI) based Decision Support System for predicting Water Quality Index (WQI) using an appropriate Machine Learning Algorithm. The project will adopt 2000 water quality instances retrieved from water monitoring stations as the target dataset. Water quality parameters that will be considered include Temperature, Dissolved Oxygen, pH, Conductivity, Biological Oxygen Demand (BOD), Nitrate + Nitrite , Fecal Coliform, and Total Coliform.

The project will calculate Water Quality Index using a published standard methodology applicable in the target dataset. The project will develop a number of machine learning models including Random Forest, XGBoost, and LightGBM. The best performing model will be interpreted using SHAP values to understand the relationship between the input parameters and the predicted WQI. The project will make recommendations on water quality based on the predicted WQI.

(b) Individual Role and Tasks

As an individual member of project team, I will primarily contribute to the **decision-support, recommendation, system integration, and testing stages** of the project. My responsibilities include:

- Study existing approaches for the water-quality decision-support and recommendation systems.
- Analyze the predicted WQI and the corresponding water-quality categories.

- Design a recommendation logic based on the predicted water-quality conditions.
- Develop a decision-support component that presents the predicted WQI, water-quality category, and appropriate recommendation information.
- Integrate model prediction results with the explanation and the recommendation components.
- Develop suitable visual outputs for communicating the water-quality conditions.
- Test the decision-support workflow using the different water-quality conditions.
- Evaluate whether the generated outputs are understandable and the consistent with the WQI categories.
- Support system integration and the final project testing.
- Contribute to the research paper writing, final documentation, and presentation.

Individual Deliverables

Literature Review Contribution
Decision-Support System Design
Water Quality Recommendation Logic
WQI Category Interpretation Module
Recommendation Module
Model Output Integration
Decision-Support Visualization
System Testing Report
Integrated Project Demonstration
Research Paper Contribution
Final Project Documentation
Presentation Slides

(c) Specific Approach

The given project will adopt the standard approach used in machine learning. First of all, the water quality data set will be collected and preprocessed. It includes removing missing values and other irrelevant information to prepare the data for training.

Then, with the use of the published method for calculating the Water Quality Index (WQI), that is applicable to the given data set, the WQI will be determined. Following that, the Exploratory Data Analysis will be performed to examine the relationship between various parameters and detect patterns in the data set.

After that, feature engineering will be applied, and various algorithms, among which Random Forest, XGBoost, and Ligther GBM, will be trained on the given data set. The performance of all the models will be compared using evaluation metrics such as Mean Absolute Error, Root Mean Square Error, and R2 Score. Then, the performance of the best model will be explained using the SHAP (SHapley Additive exPlanations). In the end, a Decision Support System will be designed that will indicate the predicted WQI and give a recommended statement on the quality of water based on the prediction.

(d) Design Phases

Phase Activities

Phase 1 Literature Review and Problem Identification
Phase 2 Dataset Collection and Data Preprocessing
Phase 3 Water Quality Index Calculation
Phase 4 Exploratory Data Analysis and Feature Engineering
Phase 5 Machine Learning Model Development
Phase 6 Model Training, Hyperparameter Tuning and Evaluation
Phase 7 Explainable AI Implementation using SHAP
Phase 8 Decision Support System Development
Phase 9 Testing, Documentation, Research Paper Preparation and Final Presentation

Outcome Matrix: Describe your plan to demonstrate each of the outcomes below.

Outcomes:	**Plan for demonstrating outcome:**

a) an ability to apply knowledge of mathematics, science, and engineering	Apply water-quality concepts, WQI classification principles, software engineering, and data interpretation to convert model predictions into meaningful water-quality categories and recommendations.
c) an ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	Design a decision-support and recommendation workflow while considering usability, environmental relevance, reliability of recommendations, computational efficiency, data limitations, and sustainability.
d) an ability to function on multidisciplinary teams	Collaborate with team members by integrating outputs from data processing, WQI prediction, and explainability components into a unified decision-support workflow.
e) an ability to identify, formulate, and solve engineering problems	Solve the problem of converting numerical WQI predictions into understandable information and recommendations that can support interpretation of water-quality conditions.
g) an ability to communicate effectively	Present system outputs, WQI categories, recommendations, workflow diagrams, testing results, project documentation, and demonstrations clearly to the team and faculty.
k) an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice	Use Python, data-processing libraries, visualization tools, machine-learning integration tools, and software development tools to develop, test, and demonstrate the decision-support component.

Realistic Constraints:

There are various realistic constraints taken into consideration in the project including lack or imperfection of water quality readings, variance in water quality at various monitoring stations, computing constraints during model training, and availability of certain water quality parameters only for calculation of the Water Quality Index. Environmental sustainability, proper use of open-source data and the generation of reliable recommendations is also taken into account.

Engineering Standards:

The project will comply with the established engineering and research standards throughout its implementation. Standard machine learning techniques like data pre-processing, feature engineering, validation of models, hyperparameter tuning and evaluation of the models with MAE, RMSE and R^2 Score will be used. Explainable Artificial Intelligence (XAI) will be utilized via SHAP. The Decision Support System will provide recommendations on the basis of WQI values according to certain water quality categories. Documentation and research reporting will be done according to IEEE standards.